



Forces, Friction and Inclined Planes

• Worksheet 1

Mechanics Chapter 6: Forces, Friction and Inclined Planes. Answer all questions. Show your working clearly. Total: 86 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. Use $F \leq \mu R$, with $F = \mu R$ only when friction is limiting or the body is sliding. Give answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Resolving forces	19	
B Coefficient of friction and limiting equilibrium	20	
C Bodies on inclined planes	21	
D Connected particles with friction	21	
Roar-level challenge	5	
Total	86	

Section A Resolving forces

19 marks

1. A force of 10 N acts at 30° above the horizontal. Find its horizontal and vertical components. [2]

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2. Two forces act on a particle: 6 N due east, and 8 N in the direction 60° north of east. [3]
Find the magnitude of the resultant force and the angle it makes with the easterly direction.

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3. A particle is in equilibrium under three horizontal forces: 5 N due east, 12 N due south, and a force P N acting at an angle θ north of west. [3]
Find P and θ .

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4. A particle of weight 20 N hangs in equilibrium from two light strings. The strings make angles of 30° and 60° with the horizontal. [4]
Find the tension in each string.

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5. A particle of mass 2 kg hangs from a light string. It is held in equilibrium by a horizontal force P N, with the string at 25° to the vertical.

(a) Draw a force diagram for the particle.

[1]

(b) Find the tension in the string.

[3]

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(c) Find P .

[2]

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(d) State what happens to the angle of the string if P is increased.

[1]

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(Total for Question 5 is 7 marks)



Section B Coefficient of friction and limiting equilibrium

20 marks

6. A box of mass 5 kg rests on rough horizontal ground. The coefficient of friction is 0.3. Find the maximum frictional force. [2]

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7. A box of mass 10 kg rests on rough horizontal ground, with $\mu = 0.4$. A horizontal force of 30 N is applied. [3]
Determine whether the box moves, and find the frictional force acting.

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8. A box of mass 4 kg is pulled along rough horizontal ground by a horizontal force of 30 N, with $\mu = 0.25$. [3]
Find the acceleration of the box.

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9. A box of mass 8 kg rests on rough horizontal ground. It is pulled by a force of 40 N at 30° above the horizontal, and is on the point of sliding. [4]
Find the coefficient of friction.

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10. A particle of mass 6 kg rests on rough horizontal ground, with coefficient of friction 0.35. A force of magnitude P N is applied, acting downwards at 20° below the horizontal.

(a) Given that the particle is in limiting equilibrium, find P . [4]

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(b) The force is increased to 40 N, at the same angle. Find the acceleration of the particle. [4]

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(Total for Question 10 is 8 marks)

**Section C Bodies on inclined planes**

21 marks

11. A particle of mass 2 kg slides down a smooth plane inclined at 30° to the horizontal. Find its acceleration. [2]

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12. A particle of mass 5 kg is held at rest on a smooth plane inclined at 20° , by a force acting up the line of greatest slope. [3]
Find the force and the normal reaction.

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13. A particle of mass 3 kg slides down a rough plane inclined at 25° , with $\mu = 0.3$. Find its acceleration. [3]

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14. A particle rests on a rough plane inclined at angle α and is on the point of sliding down. [4]

(a) Show that $\mu = \tan \alpha$.

(b) Hence find the least value of μ for which a particle can rest on a plane inclined at 35° .

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15. A particle of mass 4 kg is projected up a line of greatest slope of a rough plane inclined at 30° to the horizontal, with initial speed 8 ms^{-1} . The coefficient of friction is 0.2.

(a) Find the deceleration of the particle as it moves up the plane. [3]

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(b) Find the distance it travels up the plane before coming to rest. [2]

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(c) Determine whether the particle stays at rest or slides back down. [2]

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(d) If it slides back, find its acceleration as it moves down the plane. [2]

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(Total for Question 15 is 9 marks)



Section D Connected particles with friction

21 marks

16. A particle of mass 3 kg on a rough horizontal table ($\mu = 0.2$) is joined by a light inextensible string over a smooth pulley at the edge of the table to a particle of mass 2 kg hanging freely. The system is released from rest. [2]
Find the frictional force on the 3 kg particle.

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17. For the system in Question 16, find the acceleration and the tension in the string. [3]

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18. Boxes A (4 kg) and B (6 kg) lie on rough horizontal ground ($\mu = 0.25$), joined by a light inextensible rope. A horizontal force of 40 N pulls B directly away from A . [3]
Find the acceleration and the tension in the rope.

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19. A particle of mass 3 kg lies on a rough plane inclined at 30° ($\mu = 0.2$). It is joined by a light inextensible string, passing over a smooth pulley at the top of the plane, to a particle of mass 4 kg hanging freely. [4]

Find the acceleration of the system and the tension in the string.

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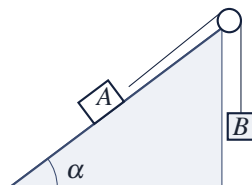
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20. A particle A of mass 5 kg lies on a rough plane inclined at angle α , where $\tan \alpha = \frac{3}{4}$. The coefficient of friction between A and the plane is 0.5 . A is joined by a light inextensible string, passing over a smooth pulley at the top of the plane, to a particle B of mass $m\text{ kg}$ hanging freely.



- (a) Given that A is on the point of sliding up the plane, find m . [4]

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- (b) Given instead that A is on the point of sliding down the plane, find m . [2]

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- (c) In the case $m = 8$, the system is released from rest. Find the acceleration of A . [3]

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(Total for Question 20 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. A box of mass 10 kg rests on rough horizontal ground, with coefficient of friction 0.75. A force P N, [5]
at an angle θ above the horizontal, is just enough to make the box move.

Show that $P = \frac{73.5}{\cos \theta + 0.75 \sin \theta}$, and find the least possible value of P and the angle at which it acts.

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END OF WORKSHEET

Total: 86 marks